CAM – EXAM 21/06/2021

SOLUTIONS TO NUMERICAL PROBLEMS

METAL FORMING

A cylindrical sample, with an initial height $h_0 = 500$ mm and an initial diameter $d_0 = 400$ mm, is hot forged between two flat dies. The material behaves like a perfectly plastic material with $Y = 320\ \mathrm{MPa}$.

1) Neglecting friction, determine the ideal work required to reach a true strain equal to $\varepsilon = -0{,}05$.
2) Neglecting friction, determine the final height of the sample if the true strain is equal to $\varepsilon = -0{,}06$.
3) Neglecting friction, determine the maximum sample diameter that can be obtained with a press capable of achieving a maximum force equal to 50 MN (a 50 MN press).
4) Assuming that the friction is not negligible and is equal to $\mu = 0{,}15$, determine the force to obtain a final diameter equal to $d_1 = 430\ \mathrm{mm}$. Check if the open die forging operation can be done with a 65 MN press.

Solution

1) Neglecting friction, determine the ideal work required to reach a true strain equal to $\varepsilon = -0{,}05$.

The ideal work required is equal to:

$$W = uV$$

where V is the volume of the cylindrical sample

$$V = \frac{\pi d_0^2}{4} h_0 = \frac{\pi \cdot 400^2}{4} \cdot 500 = 62.831.853\ \mathrm{mm}^3$$

and u is the strain energy density

$$u = |Y\epsilon| = 320 \cdot 0{,}06 = 16\ \mathrm{MPa} = 0{,}016 \frac{\mathrm{J}}{\mathrm{mm}^3}$$

Therere, it follows

$$W = 0{,}016 \cdot\ 62.831.853 = 1.005.310\ \mathrm{J} = 1{,}01\ \mathrm{MJ}$$

2) Neglecting friction, determine the final height of the sample if the true strain is equal to $\varepsilon = -0{,}06$.

The material behaves like a perfectly plastic material, therefore there is no elastic recovery. It follows that the final true strain is $\varepsilon = -0.06$.

Recalling that

$$\varepsilon = \ln \frac{h_1}{h_0}$$

The final height of the sample is

$$h_1 = h_0 e^{\varepsilon} = 500\ e^{-0{,}06} = 470{,}88\ mm$$

3) Neglecting friction, determine the maximum sample diameter that can be obtained with a press capable of achieving a maximum force equal to 50 MN (a 50 MN press).

The maximum force is equal to $F_{max} = Y \cdot A_{max} = Y \cdot \frac{\pi}{4} d_{max}^2$

Therefore, the maximum diameter is equal to

$$d_{max} = \sqrt{\frac{4 \cdot F_{max}}{\pi \cdot Y}} = \sqrt{\frac{4 \cdot\ 50.000.000}{\pi \cdot 320}} = 446{,}031\ \mathrm{mm}$$

4) Assuming that the friction is not negligible and is equal to $\mu = 0{,}15$, determine the force to obtain a final diameter equal to $d_1 = 430$ mm. Check if the open die forging operation can be done with a 65 MN press.

The force can be computed as

$$F = p_{av}\frac{\pi d_1^2}{4} = Y\left(1 + \frac{2\mu r_1}{3h_1}\right)\cdot\frac{\pi d_1^2}{4}$$

where h_1 can be computed considering the volume of the sample constant

$$h_1 = \frac{d_0^2}{d_1^2}h_0 = \frac{400^2}{430^2}\cdot 500 = 432{,}666\ \mathrm{mm}$$

Ne segue:

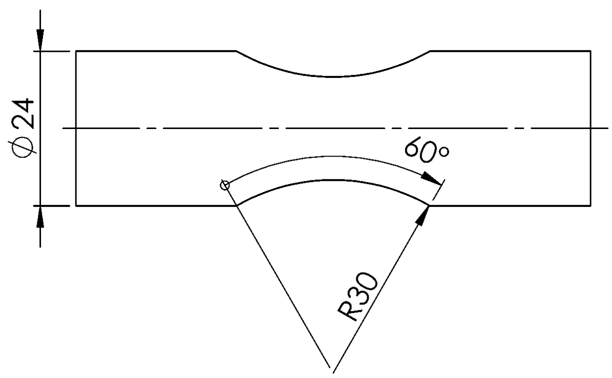
$$F = 320\left(1 + \frac{0{,}15\cdot 430}{3\cdot 432{,}666}\right)\cdot\frac{\pi}{4}430^2 = 48.779.643\ \mathrm{N} = 48{,}8\ \mathrm{MN} < 65\ \mathrm{MN}$$

The forging operation can be done on a 65 MN press.

TURNING

In the figure, the final geometry of a mechanical component is represented. In order to reach that shape, an external finishing operation is made on a lathe, mounting the workpiece between centers, and considering the following data: $a_p = 1$ mm, $k_{cs} = 2100$ MPa, $x = 0{,}29$; $\kappa_r = 65°$, $\kappa_r' = 65°$; component length equal to 90 mm.

1) Considering $n = 3200\ RPM$, determine the maximum and the minimum cutting speeds.
2) Considering $n = 3200\ RPM$ and $f = 0{,}15$ mm/rev., determine the cutting force and power to machine the external cylindrical surface.
3) The company estimates for each machined component a tool contact time of 40 s and a loading / unloading time of 30 s. It is planned to use tool inserts with two cutting edges each, which guarantee (under current cutting conditions) a tool life per cutting edge equal to 4,5 minutes. The tool (cutting edge) changing time is equal to 30 s. The cutting edge cannot be replaced during a component machining and, at the end of the shift, the last insert mounted on the tool body is discarded even if it has a residual life sufficient for other machining operations or a completely new cutting edge. Calculate the production time per part (T_p), the number of parts that can be machined in an 8-hour shift, and the total cost of tool inserts knowing that the unit cost is 8 €/insert.



Solution

1) Considering $n = 3200\ RPM$, determine the maximum and the minimum cutting speeds.

The machining operation is done considering a constant revolution per minute; therefore, the maximum cutting speed is the cutting speed at the maximum final diameter, equal to $D_{max} = 24\ mm$.

$$v_{c,max} = \frac{n\cdot\pi D_{max}}{1000} = \frac{3200\cdot\pi\cdot 24}{1000} = 241{,}27\ \mathrm{m/min}$$

Similarly, the minimum cutting speed is the cutting speed at the minimum final diameter, equal to

$$D_{min} = D_{max} - 2\cdot R\cdot\left(1 - \cos{}^{\alpha}/_{2}\right) = 24 - 2\cdot 30\cdot\left(1 - \cos{}^{60°}/_{2}\right) = 15{,}96\ \mathrm{mm}$$

Therefore,

$$v_{c,min} = \frac{n \cdot \pi D_{min}}{1000} = \frac{3200 \cdot \pi \cdot 15{,}96}{1000} = 160{,}45 \text{ m/min}$$

2) *Considering $n = 3200\ RPM$ and $f = 0{,}15\ mm/rev.$, determine the cutting force and power to machine the external cylindrical surface.*

The cutting force is equal to

$$F_c = \frac{k_{cs}}{(sin\,\kappa_r)^x} \cdot f^{1-x} \cdot a_p = \frac{2100}{(sin\,65°)^{0{,}29}} \cdot 0{,}15^{1-0{,}29} \cdot 1 = 561{,}87 \text{ N}$$

Therefore, the cutting power is equal to

$$P_c = \frac{F_c \cdot v_{c,max}}{60} = \frac{561{,}87 \cdot 241{,}27}{60} = 2.259{,}4 \text{ W} = 2{,}3 \text{ kW}$$

3) *The company estimates for each machined component a tool contact time of 40 s and a loading / unloading time of 30 s. It is planned to use tool inserts with two cutting edges each, which guarantee (under current cutting conditions) a tool life per cutting edge equal to 4,5 minutes. The tool (cutting edge) changing time is equal to 30 s. The cutting edge cannot be replaced during a component machining and, at the end of the shift, the last insert mounted on the tool body is discarded even if it has a residual life sufficient for other machining operations or a completely new cutting edge. Calculate the production time per part (T_p), the number of parts that can be machined in an 8-hour shift, and the total cost of tool inserts knowing that the unit cost is 8 €/insert.*

The production time per part $(\mathrm{T_p})$ is equal to

$$T_p = T_h + T_m + \frac{T_t}{n_p}$$

where n_p is the number of parts a cutting edge is able to machine, equal to

$$n_p = \left\lfloor \frac{T}{T_m} \right\rfloor = \left\lfloor \frac{4{,}5 \cdot 60}{40} \right\rfloor = \lfloor 6{,}75 \rfloor = 6$$

Therefore,

$$T_p = T_h + T_m + \frac{T_t}{n_p} = 30 + 40 + \frac{30}{6} = 75 \text{ s}$$

The number of parts that can be machined in an 8-hour shift is

$$n_{parts} = \left\lfloor \frac{T_{shift}}{T_p} \right\rfloor = \left\lfloor \frac{8 \cdot 3600}{75} \right\rfloor = 384 \text{ parts/shift}$$

The number of cutting edges to be used in a shift production is

$$n_{cutting\ edges} = \left\lceil \frac{n_{parts}}{n_p} \right\rceil = \left\lceil \frac{384}{6} \right\rceil = 64$$

Considering two cutting edges per tool insert, the required number of tool inserts per shift is equal to

$$n_{inserts} = \left\lceil \frac{n_{cutting\ edges}}{2} \right\rceil = \left\lceil \frac{64}{2} \right\rceil = 32$$

Therefore, the total cost of tool inserts is

$$C_{inserts} = n_{inserts} \cdot c_{insert} = 32 \cdot 8 = 256 \text{ €}$$

DRILLING

Consider a machining cycle to obtain a through hole with a diameter of 40 mm, a depth of 80 mm and a tolerance grade equal to IT7. The machining cycle involves a drilling operation with a twist drill bit with a diameter equal to 37 mm and a reaming operation. The machining operations are carried out on a machine tool with a nominal power equal to 15 kW and an efficiency equal to 90%. Determine the cutting torque, the cutting power, and the machining time for the reaming operation. Is the reaming operation feasible on the selected machine tool?

Reaming data:

- Cutting speed: $v_c = 120$ m/min
- Feed: $f = 1$ mm/revolution
- Number of cutting edges: Z = 8
- Cutting pressure: $k_c = 2100$ MPa
- Extra path in and out: 2 mm each

Solution

1) Cutting torque

In the reaming operation each cutting edge will remove material with a depth of cut equal to

$$a_p = \frac{D_{final} - D_{drilling}}{2} = \frac{40 - 37}{2} = 1{,}5 \text{ mm}$$

It follows a chip area before deformation equal to

$$A_D = f_z \cdot a_p = \frac{f}{Z} \cdot a_p = \frac{1}{8} \cdot 1{,}5 = 0{,}1875 \text{ mm}^2$$

and a cutting force per each cutting edge equal to

$$F_c = k_c A_D = 2100 \cdot 0{,}1875 = 393{,}75 \text{ N}$$

In a reaming operation all the cutting edges are simultaneously working with identical cutting parameters, therefore

$$T_c = Z \cdot F_c \cdot \frac{D_{final} + D_{drilling}}{4} \cdot \frac{1}{1000} = 8 \cdot 393{,}75 \cdot \frac{40 + 37}{4} \cdot \frac{1}{1000} = 60{,}64 \text{ Nm}$$

2) Cutting power

Therefore, the cutting power is equal to

$$P_c = T_c \omega$$

$$\omega = \frac{v_c}{D_f/2} \cdot \frac{1000}{60} = \frac{120}{40/2} \cdot \frac{1000}{60} = 100 \ \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 60{,}64 \cdot 100 = 6.064 \text{ W} = 6{,}1 \text{ kW}$$

3) Machining time

The machining time is equal to

$$t_m = \frac{L_m}{v_f} = \frac{H + e_{in} + e_{out}}{f \cdot n}$$

where

$$n = \frac{1000 \cdot v_c}{\pi D_f} = \frac{1000 \cdot 120}{\pi \cdot 40} = 955 \text{ RPM}$$

Therefore

$$t_m = \frac{80 + 2 + 2}{1 \cdot 955} \cdot 60 = 5{,}3 \text{ s}$$

4) Is the reaming operation feasible on the selected machine tool?

The power limit is $P_{c,limit} = E \cdot P_e = 0{,}9 \cdot 15 = 13{,}5 \text{ kW}$, therefore the reaming operation is feasible.